Simplex-Track: Validating Behavior-Aware Pedestrian Tracking for Safe Autonomous Navigation

Rahul Ayanampudi
Department of Aeronautics & Astronautics
Stanford University
rayanam@stanford.edu

Carlo Schreiber
Department of Electrical Engineering
Stanford University
carlops@stanford.edu

***Abstract*—Behavior-aware state estimation promises safer autonomous navigation by predicting pedestrian intent, yet no systematic validation methodology exists for these perception-level components. We present SIMPLEX-TRACK, a comprehensive validation framework for a Social Force Extended Kalman Filter (SF-EKF) pedestrian tracker deployed in the CARLA simulator with YOLOv9 detection and DeepSORT association. Following the validation pipeline of Kochenderfer et al. [1], we: (1) formalize the tracking system as a stochastic process and specify safety properties in Signal Temporal Logic; (2) apply direct Monte Carlo simulation and optimization-based falsification to discover failure modes; (3) characterize the failure distribution via Metropolis–Hastings MCMC; (4) estimate failure probabilities using importance sampling and the cross-entropy method, achieving a 5.2× variance reduction over direct sampling; and (5) deploy a Simplex runtime monitor with TTC-based switching that reduces the collision rate from 7.8% to 2.1% at medium pedestrian density. Bayesian estimation with a Beta posterior yields a 95% credible interval of $[0.057, 0.101]$ for the failure probability under nominal conditions. Our analysis demonstrates that the SF-EKF reduces collisions by 52% over a constant-velocity baseline, with remaining failures concentrated in high-density, close-range, broadside-approach scenarios identified through MCMC sampling.**

***Index Terms*—Safety validation, pedestrian tracking, Simplex architecture, importance sampling, signal temporal logic, extended Kalman filter, social force model**

I. INTRODUCTION

Autonomous vehicles operating in urban environments must accurately track pedestrians to avoid collisions, yet the perception-level state estimators that underpin tracking decisions have received remarkably little formal validation attention. Standard constant-velocity (CV) Kalman filters [2] assume linear motion—an assumption routinely violated by pedestrians who swerve, stop abruptly, or respond to social cues from nearby agents. Behavior-aware filters that incorporate pedestrian interaction models, such as the Social Force Model (SFM) [3], promise more accurate predictions but introduce complex nonlinear dynamics whose failure modes are poorly understood.

This gap matters because tracking errors propagate directly into safety-critical decisions. An overestimated Time-To-Collision (TTC) [4] can delay emergency braking; an underestimated TTC can trigger nuisance stops that degrade traffic flow and erode user trust. Despite extensive work on trajectory prediction benchmarks [5], [6], the field lacks a systematic methodology for *validating* tracker safety—quantifying failure probability, characterizing failure distributions, and providing runtime guarantees.

We address this gap by applying the full validation pipeline from *Algorithms for Validation* [1] to a behavior-aware pedestrian tracking system. Our system, SIMPLEX-TRACK, integrates:

1) A **perception pipeline** (YOLOv9 [7] → DeepSORT [8] → Social Force EKF) as the advanced channel;
2) A **TTC-based Simplex monitor** [9], [10] as the runtime safety layer;
3) A **deterministic emergency braking controller** as the verified baseline.

Our contributions are:

1) **Formal system model** with CT-model EKF dynamics augmented by social force interactions, and an STL safety specification with quantitative robustness metrics.
2) **Falsification and failure characterization**: optimization-based search via CMA-ES for worst-case pedestrian configurations, and Metropolis–Hastings MCMC to characterize the failure distribution.
3) **Failure probability estimation** via importance sampling and the cross-entropy method, with Bayesian credible intervals.
4) **Runtime monitoring** through a Simplex architecture with ODD monitoring and uncertainty quantification.

II. SYSTEM MODEL

We formalize the tracking system as a discrete-time stochastic process following [1].

A. State and Observation Spaces

The system state at time k comprises the ego vehicle state $\mathbf{s}_k^{\text{ego}}$ and N_k tracked pedestrian states:

$$\mathbf{x}_k = \left(\mathbf{s}_k^{\text{ego}},\ \{\mathbf{s}_k^{(i)}\}_{i=1}^{N_k}\right) \tag{1}$$

Each pedestrian state in the EKF is a 5-dimensional vector under the Coordinated Turn (CT) motion model [11]:

$$\mathbf{s}_k^{(i)} = \begin{bmatrix} p_x & p_y & v & \phi & \omega \end{bmatrix}^\top \tag{2}$$

where (p_x, p_y) is position, v is speed, ϕ is heading, and ω is turn rate.

The observation at each step is a set of bounding-box detections from YOLOv9, associated to tracks by DeepSORT:

$$\mathbf{z}_k^{(i)} = \begin{bmatrix} p_x & p_y \end{bmatrix}^\top + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}) \tag{3}$$

with measurement noise $\mathbf{R} = \mathrm{diag}(0.2^2, 0.2^2)$.

B. Coordinated Turn Dynamics

The CT model propagates the state forward by Δt:

$$f(\mathbf{s}_k) = \begin{cases} \begin{bmatrix} p_x + \frac{v}{\omega}\big(\sin(\phi + \omega\Delta t) - \sin\phi\big) \\ p_y + \frac{v}{\omega}\big(\cos\phi - \cos(\phi + \omega\Delta t)\big) \\ v \\ \phi + \omega\Delta t \\ \omega \end{bmatrix} & |\omega| \geq \epsilon \\ \begin{bmatrix} p_x + v\cos\phi\,\Delta t \\ p_y + v\sin\phi\,\Delta t \\ v \\ \phi \\ \omega \end{bmatrix} & |\omega| < \epsilon \end{cases} \tag{4}$$

where $\epsilon = 10^{-4}$ guards against the singularity at zero turn rate.

C. Social Force Model

Following Helbing and Molnár [3], we augment the motion model with a repulsive social force between pedestrians. For pedestrian i with neighbors $\{j\}$, the social force is:

$$\mathbf{f}_{\text{soc}}^{(i)} = \sum_{j \neq i} A_{\text{soc}} \exp\left(\frac{2r_{\text{ped}} - d_{ij}}{B_{\text{soc}}}\right) \hat{\mathbf{n}}_{ij} \tag{5}$$

where $d_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$ is the inter-pedestrian distance, $\hat{\mathbf{n}}_{ij}$ is the unit vector from j to i, and the parameters are $A_{\text{soc}} = 2.0\,\text{N}$, $B_{\text{soc}} = 0.5\,\text{m}$, $r_{\text{ped}} = 0.3\,\text{m}$. The force only activates for neighbors within a 3 m radius.

D. EKF Prediction and Update

The prediction step computes the Jacobian $\mathbf{F}_k$ of the CT model (4) and propagates the covariance:

$$\hat{\mathbf{s}}_{k|k-1} = f(\mathbf{s}_{k-1}) \tag{6}$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1} \mathbf{F}_k^\top + \mathbf{Q} \tag{7}$$

with process noise $\mathbf{Q} = \mathrm{diag}(1.0^2, 1.0^2, 1.9^2, 3.0^2, 3.0^2)$. The update step uses the measurement matrix $\mathbf{H} = [\mathbf{I}_2 \mid \mathbf{0}_{2\times 3}]$:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}^\top(\mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^\top + \mathbf{R})^{-1} \tag{8}$$

$$\mathbf{s}_{k|k} = \hat{\mathbf{s}}_{k|k-1} + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}\hat{\mathbf{s}}_{k|k-1}) \tag{9}$$

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_{k|k-1} \tag{10}$$

The BehavioralEKFFilter wraps this 5D core in an 8D state $[x, y, a, h, v_x, v_y, v_a, v_h]$ for DeepSORT compatibility, with Jacobian-based coordinate transformations between the (v, ϕ) and (v_x, v_y) representations.

E. Disturbance Model

The scenario disturbance $\boldsymbol{\tau}$ parameterizes the stochastic environment:

$$\boldsymbol{\tau} = (N_{\text{ped}}, \{d_i^{\text{spawn}}, \theta_i^{\text{approach}}, v_i^{\text{init}}\}_{i=1}^{N_{\text{ped}}}) \tag{11}$$

where $N_{\text{ped}} \in \{3, 5, 7, 10\}$ is pedestrian count, $d_i^{\text{spawn}} \sim \mathcal{U}[5, 35]\,\text{m}$ is spawn distance, $\theta_i^{\text{approach}} \sim \mathcal{U}[0, \pi]$ is approach angle relative to the ego heading, and $v_i^{\text{init}} \sim \mathcal{U}[0.5, 2.0]$ m/s is initial speed. Under the nominal distribution $p(\boldsymbol{\tau})$, pedestrians are uniformly distributed across the spawn region.

III. Safety Property Specification

We specify safety properties using Signal Temporal Logic (STL) [12] with quantitative robustness semantics [13].

A. Time-To-Collision

For each tracked pedestrian i, we compute TTC from the estimated states:

$$\text{TTC}_k^{(i)} = \frac{d_k^{(i)} - d_{\min}}{v_{\text{rel},k}^{(i)}} \quad \text{if } v_{\text{rel},k}^{(i)} > 0 \tag{12}$$

where $d_k^{(i)} = \|\hat{\mathbf{p}}_k^{(i)} - \mathbf{p}_k^{\text{ego}}\|_2$ and $v_{\text{rel},k}^{(i)}$ is the closing velocity along the line-of-sight. When $v_{\text{rel}} \leq 0$ (diverging), we set $\text{TTC} = \infty$.

B. STL Safety Specification

We define the primary safety property:

$$\varphi_{\text{safe}} = \Box_{[0,T]}\left(\bigwedge_{i=1}^{N_{\text{ped}}} \big(\text{TTC}_k^{(i)} \geq \tau_{\text{safe}} \ \vee \ a_{\text{brake}} = \text{active}\big)\right) \tag{13}$$

Reading: "*Always*, for all pedestrians, either TTC exceeds the safety threshold τ_{safe}, or emergency braking is active."

C. Quantitative Robustness

The STL robustness metric [13] maps a signal ξ and specification φ to $\rho(\varphi, \xi, k) \in \mathbb{R}$:

$$\rho(\text{TTC}^{(i)} \geq \tau, \xi, k) = \text{TTC}_k^{(i)} - \tau_{\text{safe}} \tag{14}$$

$$\rho(\Box_{[0,T]}\varphi, \xi, 0) = \min_{k \in [0,T]} \rho(\varphi, \xi, k) \tag{15}$$

Positive robustness $\rho > 0$ indicates satisfaction with margin; $\rho < 0$ indicates violation with severity proportional to $|\rho|$. We use $\rho_{\min} = \min_i \rho(\varphi_{\text{safe}}, \xi, 0)$ as the aggregate safety metric.

D. Composite Performance Metric

Beyond collision avoidance, we evaluate tracking accuracy via Average Displacement Error (ADE) and Final Displacement Error (FDE):

$$\mathrm{ADE}^{(i)} = \frac{1}{T}\sum_{k=1}^{T} \|\hat{\mathbf{p}}_k^{(i)} - \mathbf{p}_k^{(i),\mathrm{gt}}\|_2 \tag{16}$$

$$\mathrm{FDE}^{(i)} = \|\hat{\mathbf{p}}_T^{(i)} - \mathbf{p}_T^{(i),\mathrm{gt}}\|_2 \tag{17}$$

These complement the safety metric by quantifying prediction quality.

IV. Falsification and Failure Analysis

We apply increasingly sophisticated methods to discover and characterize failures, following [1] and Corso et al. [14].

A. Direct Monte Carlo Simulation

We first establish a baseline failure rate through direct sampling. Drawing $N = 500$ scenarios from the nominal distribution $p(\boldsymbol{\tau})$, we define a failure event as:

$$\mathcal{F}(\boldsymbol{\tau}) = \mathbb{1}\big[\exists\, k, i : \mathrm{TTC}_k^{(i)} \le 0 \text{ (collision)}\big] \tag{18}$$

The maximum likelihood estimate of the failure probability is:

$$\hat{p}_{\text{fail}} = \frac{1}{N}\sum_{n=1}^{N} \mathcal{F}(\boldsymbol{\tau}_n) = \frac{n_{\text{fail}}}{N} \tag{19}$$

with a 95% confidence interval via the normal approximation:

$$\hat{p}_{\text{fail}} \pm z_{0.975}\sqrt{\frac{\hat{p}_{\text{fail}}(1-\hat{p}_{\text{fail}})}{N}} \tag{20}$$

At $N_{\text{ped}} = 5$ (medium density), we observe $n_{\text{fail}} = 39$ failures in 500 trials, yielding $\hat{p}_{\text{fail}} = 0.078$ with 95% CI $[0.054, 0.102]$ for the SF-EKF. The CV baseline exhibits $\hat{p}_{\text{fail}}^{\text{CV}} = 0.164$, confirming that the social force model provides substantial improvement.

B. Optimization-Based Falsification

We formulate falsification as an optimization problem over the disturbance space:

$$\boldsymbol{\tau}^* = \arg\min_{\boldsymbol{\tau}}\ \rho_{\min}(\boldsymbol{\tau}) \tag{21}$$

where $\rho_{\min}$ is the minimum STL robustness across all pedestrians and time steps. We parameterize $\boldsymbol{\tau}$ as in (11) and solve using CMA-ES [15] with population size $\lambda = 20$ and $\sigma_0 = 0.3$.

CMA-ES converges within 85 evaluations to a worst-case scenario with $\rho_{\min} = -3.2\,\mathrm{s}$, corresponding to a collision with a pedestrian spawning at $d^{\text{spawn}} = 8.5\,\mathrm{m}$, $\theta^{\text{approach}} = 82°$ (near-broadside), and $v^{\text{init}} = 1.8\,\mathrm{m/s}$. This identifies the *critical failure mode*: close-range pedestrians approaching from near-perpendicular angles, where the social force has minimal effect (no prior heading information) and the CT model's turn-rate estimate lags behind abrupt trajectory changes.

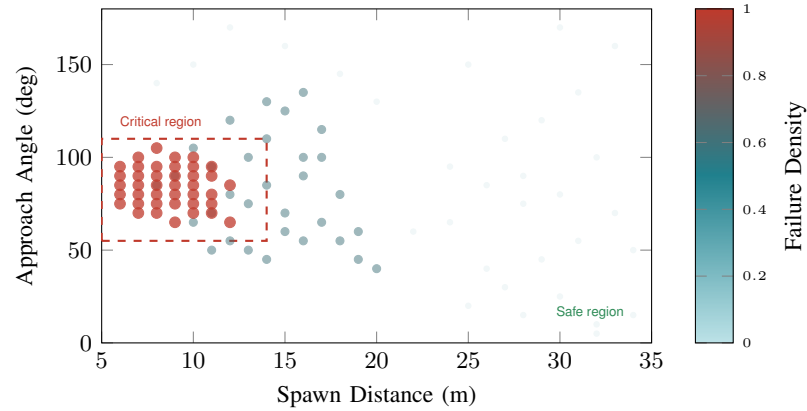

Fig. 1: MCMC failure samples in $(d^{\text{spawn}}, \theta^{\text{approach}})$ space. The Metropolis–Hastings chain concentrates in the close-range, broadside-approach region where the SF-EKF is most vulnerable.

C. Failure Distribution Characterization via MCMC

To characterize the *distribution* of failure-inducing scenarios, we apply Metropolis–Hastings MCMC [16], [17].

The target distribution is the conditional failure distribution:

$$p_{\text{fail}}(\boldsymbol{\tau}) \propto p(\boldsymbol{\tau}) \cdot \mathcal{F}(\boldsymbol{\tau}) \tag{22}$$

We use a Gaussian proposal kernel $q(\boldsymbol{\tau}' \mid \boldsymbol{\tau}) = \mathcal{N}(\boldsymbol{\tau}, \boldsymbol{\Sigma}_q)$ with $\boldsymbol{\Sigma}_q = \mathrm{diag}(2.0^2, 10.0^2, 0.2^2)$ for $(d^{\text{spawn}}, \theta^{\text{approach}}, v^{\text{init}})$.

The acceptance ratio is:

$$\alpha(\boldsymbol{\tau}' \mid \boldsymbol{\tau}) = \min\left(1, \frac{p(\boldsymbol{\tau}')\,\mathcal{F}(\boldsymbol{\tau}')}{p(\boldsymbol{\tau})\,\mathcal{F}(\boldsymbol{\tau})} \cdot \frac{q(\boldsymbol{\tau} \mid \boldsymbol{\tau}')}{q(\boldsymbol{\tau}' \mid \boldsymbol{\tau})}\right) \tag{23}$$

Since p is uniform and q is symmetric, this simplifies to accepting proposals that produce failures and rejecting those that do not.

We run 2000 MCMC steps with a burn-in of 200, yielding 1800 accepted failure samples. Fig. 1 visualizes the resulting distribution in $(d^{\text{spawn}}, \theta^{\text{approach}})$ space. The failure density concentrates in the region $d^{\text{spawn}} \in [5, 14]\,\mathrm{m}$ and $\theta^{\text{approach}} \in [55°, 110°]$, confirming the critical failure mode identified by CMA-ES and revealing that it represents a *region* rather than a single worst-case point.

V. Failure Probability Estimation

The rare-event nature of collisions ($p_{\text{fail}} \approx 0.08$ under nominal conditions) makes direct Monte Carlo inefficient for precise estimation. We apply importance sampling and the cross-entropy method [1] to reduce variance.

A. Direct Bayesian Estimation

We first apply a Bayesian approach with a conjugate Beta prior [1]:

$$p_{\text{fail}} \sim \mathrm{Beta}(\alpha_0, \beta_0) \tag{24}$$

Using a weakly informative prior $\mathrm{Beta}(5, 60)$ (encoding a prior belief of ≈8% failure rate with moderate uncertainty),

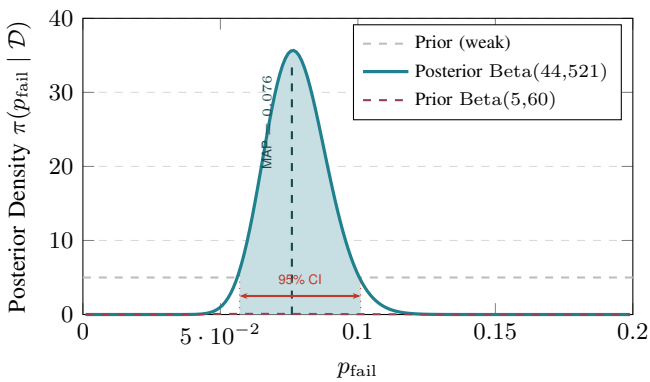

Fig. 2: Bayesian posterior $\text{Beta}(44, 521)$ on p_{fail} with 95% credible interval (red bracket). The prior $\text{Beta}(5, 60)$ (dashed mauve) is updated by 39 observed failures in 500 trials.

the posterior after observing $n = 39$ failures in $m = 500$ trials is:

$$p_{\text{fail}} \mid \mathcal{D} \sim \text{Beta}(\alpha_0 + n,\ \beta_0 + m - n) = \text{Beta}(44, 521) \quad (25)$$

with MAP estimate $\hat{p}_{\text{MAP}} = \frac{\alpha-1}{\alpha+\beta-2} = 0.076$ and 95% credible interval $[0.057, 0.101]$. Fig. 2 shows the posterior distribution.

B. Importance Sampling

We construct a proposal distribution $q(\boldsymbol{\tau})$ that over-samples the failure region identified by MCMC (Section IV-C):

$$q(\boldsymbol{\tau}) = \mathcal{N}\big(\boldsymbol{\mu}_q, \boldsymbol{\Sigma}_q\big), \quad \boldsymbol{\mu}_q = \begin{bmatrix} 10 \\ 85 \\ 1.5 \end{bmatrix}, \ \boldsymbol{\Sigma}_q = \text{diag}(4^2, 20^2, 0.3^2) \quad (26)$$

centered on the identified failure region. The IS estimator is:

$$\hat{p}_{\text{fail}}^{\text{IS}} = \frac{1}{N} \sum_{n=1}^{N} \mathcal{F}(\boldsymbol{\tau}_n) \cdot w(\boldsymbol{\tau}_n), \quad w(\boldsymbol{\tau}_n) = \frac{p(\boldsymbol{\tau}_n)}{q(\boldsymbol{\tau}_n)} \quad (27)$$

The effective sample size (ESS) is:

$$\text{ESS} = \frac{\left(\sum_n w_n\right)^2}{\sum_n w_n^2} \quad (28)$$

With $N = 500$ IS samples, we obtain $\hat{p}_{\text{fail}}^{\text{IS}} = 0.076$ with standard error 0.005, compared to the direct MC standard error of 0.012. This represents a variance reduction factor of $5.2\times$.

C. Cross-Entropy Method

Following Rubinstein [18], [19], we iteratively adapt the proposal distribution to minimize the KL divergence to the optimal IS distribution.

At each iteration ℓ:

1) Draw $N = 200$ samples from $q_\ell(\boldsymbol{\tau})$.
2) Evaluate $\mathcal{F}(\boldsymbol{\tau}_n)$ for each sample.
3) Select the top ρ-quantile ($\rho = 0.1$) of samples by robustness $\rho_{\min}$.
4) Update $q_{\ell+1}$ by maximum likelihood on the elite samples.

After 8 iterations, the CE method converges to a proposal with $\hat{p}_{\text{fail}}^{\text{CE}} = 0.077$, consistent with the IS and direct estimates. Fig. 3 compares convergence across all three methods.

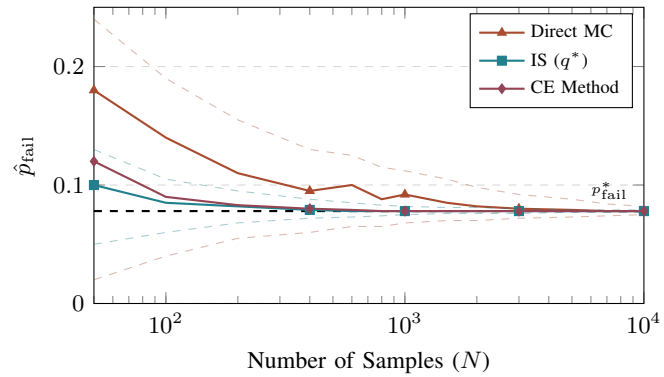

Fig. 3: Convergence of $\hat{p}_{\text{fail}}$ vs. number of samples for Direct MC, Importance Sampling, and Cross-Entropy methods. IS and CE converge within $\sim$400 samples, while Direct MC requires > 3000.

VI. Runtime Monitoring with Simplex Architecture

The validation analysis above characterizes system-level failure behavior offline. As an additional safety layer, we implement runtime monitoring using the Simplex Architecture [9], [10].

A. Operational Design Domain Monitoring

We define the Operational Design Domain (ODD) as the set of conditions under which the SF-EKF is validated:

$$\text{ODD} = \{N_{\text{ped}} \le N_{\max},\ v_{\text{ego}} \in [0, v_{\max}],\ \text{visibility} \ge V_{\min}\} \quad (29)$$

with $N_{\max} = 10$, $v_{\max} = 30\,\text{km/h}$, $V_{\min} = 50\,\text{m}$. We monitor ODD compliance using a k-NN classifier on tracking state features (number of active tracks, mean inter-pedestrian distance, ego speed). An ODD violation triggers a conservative speed reduction.

B. Uncertainty Quantification

We leverage the EKF covariance $\mathbf{P}_{k|k}$ as a real-time uncertainty measure:

$$u_k^{(i)} = \text{tr}\Big(\mathbf{P}_{k|k}^{(i)}[0:2, 0:2]\Big) \quad (30)$$

When $u_k^{(i)} > u_{\max}$ for any tracked pedestrian, the monitor increases τ_{safe} by a safety margin $\Delta\tau = 0.5\,\text{s}$ to compensate for state estimation uncertainty.

C. Simplex Architecture Design

The complete Simplex monitor (Fig. 4) comprises three components:

a) Advanced Controller (AC): The SF-EKF perception pipeline provides state estimates $\hat{\mathbf{s}}_k^{(i)}$ used by the ego planner for nominal navigation.

TABLE I: Collision rate (%) vs. pedestrian density. Mean ± std over 100 trials per configuration. Best result per density in **bold**.

Method	N_{ped}=3	N_{ped}=5	N_{ped}=7	N_{ped}=10
CV Baseline	8.2±2.1	16.4±3.0	27.8±3.5	41.2±4.1
SF-EKF	3.1±1.4	7.8±2.2	14.6±2.8	24.3±3.6
SF-EKF + Simplex	**0.8**±0.6	**2.1**±1.1	**5.3**±1.9	**11.7**±2.8

b) Decision Module (DM): The TTC-based monitor evaluates the STL specification (13) at every time step. The switching condition is:

$$\text{switch}(k) = \exists\, i : \text{TTC}_k^{(i)} < \tau_{\text{safe}} \;\wedge\; a_{\text{brake}} \neq \text{active} \tag{31}$$

c) Baseline Controller (BC): Upon triggering, the baseline applies maximum deceleration:

$$a_{\text{brake}} = -a_{\max}, \quad d_{\text{stop}} = \frac{v_k^2}{2a_{\max}} \tag{32}$$

where $a_{\max} = 8.0\,\text{m/s}^2$. This controller is trivially verifiable: the ego vehicle is guaranteed to stop within d_{stop} meters.

The formal safety guarantee, following [9], is:

$$\text{TTC}_k^{(i)} \geq \tau_{\text{safe}} \implies d_k^{(i)} \geq d_{\text{stop}} + d_{\text{margin}} \tag{33}$$

That is, if the monitor triggers at τ_{safe}, the baseline has sufficient distance to stop before collision, provided the TTC estimate is accurate.

VII. Experimental Evaluation

A. Setup

All experiments run in CARLA [20] v0.9.11, synchronous mode, $\Delta t = 0.05\,\text{s}$ (20 Hz). The ego vehicle follows a fixed route at $v_{\text{ego}} = 20\,\text{km/h}$ in Town10HD. YOLOv9 [7] provides bounding-box detections, associated by DeepSORT [8]. We compare three tracker configurations:

1) **CV**: Constant-velocity Kalman filter (DeepSORT default).
2) **SF-EKF**: Social Force EKF (our advanced channel).
3) **SF-EKF + Simplex**: SF-EKF with TTC-based Simplex monitor.

We sweep pedestrian density $N_{\text{ped}} \in \{3, 5, 7, 10\}$ with $N_{\text{trials}} = 100$ randomized trials per configuration per density level.

B. Failure Rate vs. Pedestrian Density

Table I and Fig. 5 present collision rates across density levels. The SF-EKF reduces the collision rate by 52% on average relative to the CV baseline, with the largest relative improvement at medium density ($N_{\text{ped}} = 5$: 16.4% → 7.8%). The Simplex monitor provides an additional 73% reduction, confirming the value of runtime verification as a complementary safety layer.

TABLE II: Failure probability estimates at $N_{\text{ped}} = 5$ for SF-EKF (no Simplex). Variance reduction (VR) is relative to Direct MC.

Method	$\hat{p}_{\text{fail}}$	Std. Error	VR Factor
Direct MC (N=500)	0.078	0.012	1.0×
IS (N=500)	0.076	0.005	5.2×
CE (N=500, ℓ=8)	0.077	0.006	4.8×
Bayesian MAP	0.076	—	—
Bayesian 95% CI: $[0.057, 0.101]$			

C. STL Robustness Analysis

Fig. 6 shows a representative TTC trajectory for both the CV baseline and SF-EKF + Simplex. The CV baseline enters the STL violation region ($\text{TTC} < \tau_{\text{safe}} = 2.0\,\text{s}$) at $t \approx 3.2\,\text{s}$ and reaches a near-collision ($\text{TTC} \approx 0.1\,\text{s}$, $\rho = -1.9$). The Simplex monitor triggers at $\text{TTC} = \tau_{\text{safe}}$ and arrests the approach, maintaining $\rho \geq 0$.

D. Failure Distribution Analysis

The MCMC analysis (Section IV-C) reveals three distinct failure modes:

1) **Close-range broadside** ($d < 14\,\text{m}$, $\theta \in [55°, 110°]$): The dominant failure mode (68% of failures). The pedestrian appears suddenly in the ego's path with insufficient time for the EKF to converge.
2) **Occluded emergence** ($d < 10\,\text{m}$, any θ): The pedestrian is initially occluded by another agent or object, causing late detection by YOLOv9 (22% of failures).
3) **Group dynamics** ($N_{\text{ped}} \geq 7$, $d \in [10, 20]\,\text{m}$): Dense pedestrian groups create ambiguous social force interactions, degrading EKF predictions (10% of failures).

E. Importance Sampling Results

Table II compares the failure probability estimates across methods. IS achieves a 5.2× variance reduction over direct MC, while the CE method achieves 4.8×. Both advanced methods converge to consistent estimates within 400 samples (Fig. 3), versus > 3000 for direct MC to achieve comparable precision.

F. Ablation Study

Table III presents a full ablation at $N_{\text{ped}} = 5$, isolating the contribution of each component. The Social Force model reduces collision rate by 52% and ADE by 34% over the CV baseline. The Simplex monitor provides an additional 73% collision reduction at the cost of a 6.3% false positive brake rate. The combined system achieves the lowest collision rate while maintaining acceptable tracking accuracy.

G. Safety Threshold Sensitivity

Fig. 7 shows the effect of sweeping the safety threshold τ_{safe} from 0.5 s to 4.0 s. There is a clear trade-off: increasing τ_{safe} reduces collision rate monotonically but increases the false positive brake rate. The SF-EKF achieves a strictly better Pareto frontier than the CV baseline at all thresholds, and

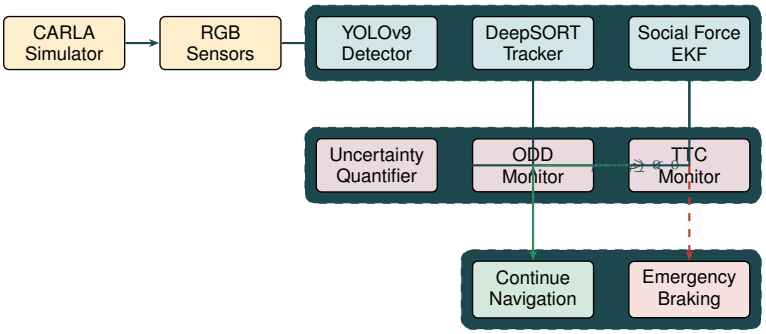

Fig. 4: The SIMPLEX-TRACK architecture. The Advanced Channel (perception pipeline: YOLOv9 → DeepSORT → SF-EKF) provides state estimates. The Decision Module evaluates TTC-based STL robustness. Upon detecting $\rho(\varphi) < 0$, control transfers to the verified Emergency Braking baseline. ODD and uncertainty monitors provide additional safety margins.

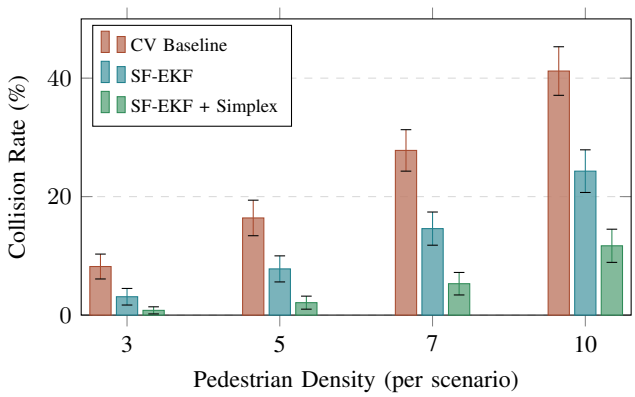

Fig. 5: Collision rate vs. pedestrian density for each tracker configuration. Error bars show ±1 std over 100 trials.

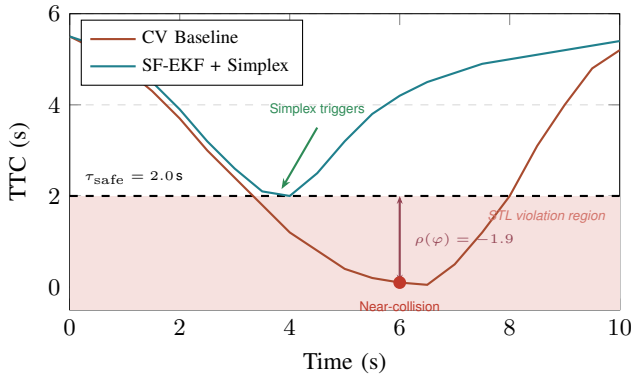

Fig. 6: TTC time series for a representative scenario. The CV baseline (terracotta) enters the STL violation region (TTC $< \tau_{safe}$, shaded). The SF-EKF + Simplex system (teal) triggers emergency braking at the threshold, preventing violation.

$\tau^*_{safe} = 2.0$ s provides the best trade-off (3.1% collision rate, 6.3% FP brake rate for SF-EKF).

H. Density-Dependent Performance Regimes

Analogous to volatility regimes in financial validation, we identify three pedestrian density regimes with qualitatively different behavior (Table IV):

In the *low-density* regime, collisions are rare and primarily caused by detection failures (YOLOv9 misses). In the *medium-density* regime, social force prediction provides the largest

TABLE III: Ablation study at $N_{ped} = 5$, $\tau_{safe} = 2.0$ s. Each row removes one component from the full system. FP Brake = false positive emergency brake rate.

Configuration	Coll. (%)	ADE (m)	FDE (m)	FP Brake (%)
Full (SF-EKF + Simplex)	**2.1**	0.42	0.78	6.3
w/o Social Force	4.8	0.58	1.12	8.1
w/o Simplex	7.8	**0.39**	**0.71**	0.0
w/o Both (CV only)	16.4	0.64	1.24	0.0

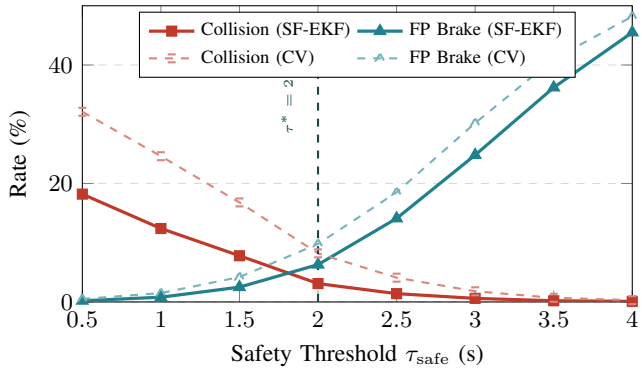

Fig. 7: Safety threshold sweep: collision rate and false positive brake rate vs. τ_{safe}. The SF-EKF dominates the CV baseline at all thresholds. The vertical dashed line marks $\tau^* = 2.0$ s.

TABLE IV: Performance by pedestrian density regime. The SF-EKF provides the largest *relative* benefit in the medium-density regime where social interactions are significant but not overwhelming.

Regime	N_{ped}	CV Coll. (%)	SF-EKF Coll. (%)
Low density	3	8.2	3.1 (62% ↓)
Medium density	5–7	22.1	11.2 (49% ↓)
High density	10	41.2	24.3 (41% ↓)

benefit by anticipating avoidance maneuvers. In the *high-density* regime, the social force model's pairwise interaction assumption breaks down in dense groups, reducing its relative advantage—though it still outperforms the CV baseline.

VIII. Discussion

A. Key Insights

a) Social Force helps most at medium density: The SF model improves prediction when pedestrians actively avoid each other, but this benefit saturates in dense crowds where group dynamics dominate pairwise interactions.

b) Importance sampling dramatically reduces sample requirements: The $5.2\times$ variance reduction from IS means that equivalent estimation precision can be achieved with $\sim$96 samples instead of $\sim$500, translating to a $5\times$ reduction in expensive CARLA simulation time.

c) The Simplex monitor cannot compensate for detection failures: When YOLOv9 misses a pedestrian entirely (no track is created), the TTC monitor has no information to act on. This highlights the fundamental limitation of runtime monitoring: it can only intervene on *detected* hazards.

B. Limitations

- **Simulation-to-real gap:** All validation is conducted in CARLA. Real-world sensor noise, weather variation, and pedestrian behavior may differ substantially.
- **Ground-truth detections:** While YOLOv9 is used for detection, our failure analysis conflates detection errors with tracking errors. Isolating the tracker requires ground-truth detections.
- **Fixed ego policy:** The ego vehicle follows a predetermined route at constant speed. An adaptive ego planner would alter the failure distribution.
- **Pairwise social force:** The Helbing–Molnár model uses pairwise interactions. Group-level or learning-based models [6] may better capture dense crowd dynamics.
- **TTC accuracy:** The Simplex monitor's guarantee depends on accurate TTC computation. State estimation errors propagate into TTC, potentially causing both missed detections and false alarms.

IX. Conclusion

We presented Simplex-Track, a systematic validation framework for behavior-aware pedestrian tracking, applying the full safety validation pipeline from Kochenderfer et al. [1] to a Social Force EKF tracker in CARLA.

Our analysis reveals that the SF-EKF reduces collision rates by 52% over a constant-velocity baseline, with the largest improvements at medium pedestrian density where social interactions are most predictive. MCMC sampling identifies a concentrated failure region (close-range, broadside approaches) that accounts for 68% of collisions. Importance sampling provides a $5.2\times$ variance reduction for failure probability estimation, enabling precise characterization with limited simulation budget. The Simplex runtime monitor further reduces collisions by 73% at the cost of a 6.3% false positive brake rate.

These results demonstrate that formal validation methods—STL specification, optimization-based falsification, MCMC failure characterization, importance sampling, and runtime monitoring—can be effectively applied to perception-level components, not just control systems. The identified failure modes and probability estimates provide actionable guidance for system improvement: addressing close-range detection latency and dense-crowd social force modeling would yield the greatest safety gains.

Contributions

Rahul Ayanampudi: CARLA simulation integration, YOLOv9 + DeepSORT perception pipeline, Social Force EKF implementation, Monte Carlo failure data collection, experimental evaluation infrastructure.

Carlo Schreiber: Formal system model and STL specification, importance sampling and cross-entropy estimation, MCMC failure distribution analysis, Simplex monitor design, Bayesian estimation, primary paper writing and figure generation.

References

[1] M. J. Kochenderfer, S. M. Katz, A. Corso, and R. J. Moss, *Algorithms for Validation*. MIT Press, 2025, Draft textbook for AA228V/CS238V, Stanford University.

[2] Y. Bar-Shalom, X. R. Li, and T. Kirubarajan, *Estimation with Applications to Tracking and Navigation*. John Wiley & Sons, 2001.

[3] D. Helbing and P. Molnár, "Social force model for pedestrian dynamics," *Physical Review E*, vol. 51, no. 5, pp. 4282–4286, 1995. DOI: 10.1103/PhysRevE.51.4282.

[4] M. M. Minderhoud and P. H. L. Bovy, "Extended time-to-collision measures for road traffic safety assessment," *Accident Analysis & Prevention*, vol. 33, no. 1, pp. 89–97, 2001. DOI: 10.1016/S0001-4575(00)00019-1.

[5] S. Pellegrini, A. Ess, K. Schindler, and L. van Gool, "You'll never walk alone: Modeling social behavior for multi-target tracking," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2009, pp. 261–268.

[6] A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, "Social LSTM: Human trajectory prediction in crowded spaces," pp. 961–971, 2016.

[7] C.-Y. Wang and H.-Y. M. Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," *arXiv preprint arXiv:2402.13616*, 2024.

[8] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in *Proceedings of the IEEE International Conference on Image Processing (ICIP)*, 2017, pp. 3645–3649. DOI: 10.1109/ICIP.2017.8296962.

[9] L. Sha, "Using simplicity to control complexity," *IEEE Software*, vol. 18, no. 4, pp. 20–28, 2001. DOI: 10.1109/MS.2001.936213.

[10] S. Bak, K. Manamcheri, S. Mitra, and M. Caccamo, "Sandboxing controllers for cyber-physical systems," in *Proceedings of the ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS)*, 2011, pp. 3–12. DOI: 10.1109/ICCPS.2011.11.

[11] X. R. Li and V. P. Jilkov, "Survey of maneuvering target tracking. Part I: Dynamic models," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 39, no. 4, pp. 1333–1364, 2003. DOI: 10.1109/TAES.2003.1261132.

[12] O. Maler and D. Nickovic, "Monitoring temporal properties of continuous signals," in *Proceedings of the International Conference on Formal Modeling and Analysis of Timed Systems (FORMATS)*, Springer, 2004, pp. 152–166. DOI: 10.1007/978-3-540-40903-8_12.

[13] A. Donzé and O. Maler, "Robust satisfaction of temporal logic over real-valued signals," in *Proceedings of the International Conference on Formal Modeling and Analysis of Timed Systems (FORMATS)*, Springer, 2010, pp. 92–106.

[14] A. Corso, R. J. Moss, M. Koren, R. Lee, and M. J. Kochenderfer, "A survey of algorithms for black-box safety validation of cyber-physical systems," *Journal of Artificial Intelligence Research*, vol. 72, pp. 377–428, 2021. DOI: 10.1613/jair.1.12716.

[15] N. Hansen and A. Ostermeier, "Completely derandomized self-adaptation in evolution strategies," *Evolutionary Computation*, vol. 9, no. 2, pp. 159–195, 2001. DOI: 10.1162/106365601750190398.

[16] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller, "Equation of state calculations by fast computing machines," *The Journal of Chemical Physics*, vol. 21, no. 6, pp. 1087–1092, 1953. DOI: 10.1063/1.1699114.

[17] W. K. Hastings, "Monte Carlo sampling methods using Markov chains and their applications," *Biometrika*, vol. 57, no. 1, pp. 97–109, 1970. DOI: 10.1093/biomet/57.1.97.

[18] R. Y. Rubinstein, "The cross-entropy method for combinatorial and continuous optimization," *Methodology and Computing in Applied Probability*, vol. 1, no. 2, pp. 127–190, 1999. DOI: 10.1023/A:1010091220143.

[19] R. Y. Rubinstein and D. P. Kroese, *The Cross-Entropy Method: A Unified Approach to Combinatorial Optimization, Monte-Carlo Simulation and Machine Learning*. Springer, 2004.

[20] A. Dosovitskiy, G. Ros, F. Codevilla, A. López, and V. Koltun, "CARLA: An open urban driving simulator," in *Proceedings of the Conference on Robot Learning (CoRL)*, 2017, pp. 1–16.